

# 8259 Programmable Interrupt Controller (PIC) — MPCA

## Module 4

**Definition:** The **Intel 8259A (PIC)** is an 8-channel programmable interrupt controller designed to manage, prioritize, and translate multiple hardware interrupt requests (**IR\_0 - IR\_7**) into 8086 vector interrupt instructions.

### 1. Detailed Technical Explanation

#### Internal Architecture & Block Diagram of 8259

##### 8259 INTERNAL ARCHITECTURE

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| Interrupt Request Register (IRR):          |
|   Latches incoming interrupt requests on pins IR0 - IR7.                        |
|                                           |
| Interrupt Mask Register (IMR):            |
|   Stores mask bits (OCW1) to selectively enable/disable IR lines                |
|                                           |
| Priority Resolver (PR):                   |
|   Determines the highest priority unmasked pending interrupt.                  |
|                                           |
| In-Service Register (ISR):               |
|   Tracks which interrupt level is currently being serviced.                   |
|                                           |
| Control Logic & Cascade Buffer:           |
|   - INT (Interrupt output to 8086 INTR pin)                                    |
|   - /INTA (Interrupt Acknowledge input from 8086)                             |
|   - CAS0, CAS1, CAS2 (Master-Slave cascading lines)                          |
|   - /SP//EN (Slave Program / Enable Buffer)                                    |
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```

### 2. 8259 Interrupt Processing Sequence

1. Peripheral asserts an interrupt line (**IR\_0 - IR\_7**).
2. **IRR** stores the request; **Priority Resolver** checks IRR against **IMR** mask and current **ISR** level.
3. 8259 sends an **INT** signal to 8086 CPU's **INTR** pin.
4. 8086 CPU responds with the **First \overline{INTA} Pulse** (Interrupt Acknowledge).
5. Upon 1st **\overline{INTA}**, the highest priority bit is set in **ISR** and cleared from **IRR**.
6. 8086 issues the **Second \overline{INTA} Pulse**. 8259 places the **8-bit Interrupt Vector Type Number (00H - FFH)** on the Data Bus (**D\_0 - D\_7**).
7. 8086 reads the vector number, multiplies by 4, and jumps to the corresponding **Interrupt Service Routine (ISR)** in the IVT table.

8. At the end of the ISR subroutine, an **End of Interrupt (EOI)** command is sent to 8259 to clear the active bit in the ISR register.
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### 3. Priority Modes in 8259

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1. **Fully Nested Mode (Default):** **IR\_0** has the highest priority and **IR\_7** has the lowest priority. Lower priority interrupts are inhibited while a higher priority ISR is running.
  2. **Automatic Rotation Mode:** Equal priority round-robin; after a channel is serviced, it is assigned the lowest priority.
  3. **Specific Rotation Mode:** Programmer manually sets which IR channel has the lowest priority.
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### 4. Must-Write Points for Exams

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- **IRR** stores requested interrupts, **ISR** stores currently executing interrupt, **IMR** masks unwanted interrupts.
  - Two INTA pulses are required from 8086 to fetch the interrupt vector from 8259.
  - A single 8259 supports 8 interrupt levels (**IR\_0 - IR\_7**); cascading in Master-Slave supports up to **64 interrupt levels**.
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### 5. Quick Recall Flow

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IR0-IR7 -> IRR -> Priority Resolver -> INT to 8086 -> 1st /INTA (Set ISR) -> 2nd /INTA (Send Vector Byte)
-> Execute ISR -> EOI Command
```